Sector Risk Model: a short summary

Sector Risk Model · short summary · code, data and full methodology (in Portuguese): github.com/fernandoxb/sector-risk-model

What it is. An end-to-end market-risk system for the 11 SPDR sector ETFs and their 110 largest holdings, built as a portfolio project for a Master's in Risk Management. It forecasts next-day and multi-day VaR and Expected Shortfall, tests those forecasts out of sample, measures how sectors and companies fall together, and simulates how a macro shock spreads from factors to sectors to single stocks. A scheduled pipeline refreshes everything every trading day; a public cockpit shows it: **sector-risk-model.streamlit.app**.

Data. Daily adjusted prices from Yahoo Finance (sector ETFs and SPY since 2005, the 110 companies since 2005), macro factors from FRED and Yahoo (10-year Treasury, 10-year breakeven inflation, high-yield and Baa credit spreads, VIX, WTI oil, dollar index). Simple returns for tail risk, log returns for GARCH. Every run is validated by a data contract (pandera) before any model runs.

1. Which VaR/ES model survives the backtest?

Four methods forecast 1-day VaR and ES using only data up to the day before, on a rolling 3-year window: historical simulation, GJR-GARCH(1,1) with Student-t innovations, Filtered Historical Simulation (FHS) and conditional EVT (GARCH plus a generalised Pareto tail; McNeil and Frey, 2000). About 46,000 forecasts per confidence level, 2008–2026, are tested with Kupiec, Christoffersen, the Acerbi–Székely Z2 test for ES (simulated p-values), the McNeil–Frey ES test and Basel traffic lights.

Series passing Kupiec, Christoffersen and Z2 at 97.5%	Model choice, 2008–2018	Confirmation, 2019–2026	109 companies (since 2008)
GARCH-EVT (official model)	**9 of 9**	**10 of 11**	**76**
GARCH-FHS	6 of 9	9 of 11	55
GJR-GARCH-t	0 of 9	0 of 11	26
Historical	0 of 9	3 of 11	8

- The method is chosen on 2008–2018 only and confirmed on 2019–2026, which the choice never saw, and on single stocks.
- The historical method reacts late to crises (clustered exceptions); GARCH-t has too thin a tail; combining GARCH volatility with the empirical or Pareto tail of the residuals fixes both.
- With a Holm correction for the 88 tests per method, no rejection of the EVT model survives, in sectors or companies.
- Where the ES is still underestimated (17 companies), the forecast is scaled by the factor that would have set Z2 to zero in the backtest (on average ×1.22).
- **Longer horizons:** 10- and 21-day VaR/ES by simulating GARCH paths. In non-overlapping 10-day blocks since 2008, 121 exceptions at 97.5% against 115 expected; the square-root-of-time rule does about as well on frequency (122), but differs day to day when volatility is far from its average.

2. Who falls with whom?

Correlation describes normal days; contagion is about bad days. For every pair of sectors the system counts how often both are in their 5% worst days, and fits a t copula (Kendall's tau, ν by maximum likelihood) to GARCH residuals (QRM, chapter 5).

- Consumer discretionary and communication services fall together on **54%** of their worst days (5% if independent). Over 10 years, with COVID-19 in the window, the average pair rises from 28% to 44%: crises are when everything falls together.
- Sectors fall together more often than they rise together (27.9% against 24.2%). A symmetric t copula on GARCH residuals simulated only 22.9%; a skewed t copula on returns, with its asymmetry set by the method of moments ($\gamma = -0.15$), simulates 26.8%.
- The cockpit's risk web links sectors by these joint falls and can switch to 1, 5 or 10-year windows.

3. What moves each sector, and each company?

- Weekly regressions of each sector on six macro factors with Newey–West errors: 34 of 66 links are significant one factor at a time; with all six factors in one regression, 28 survive and 12 vanish (rates → energy was really oil and inflation).
- Each company's daily variance splits into market, sector and own risk: median 13%, 21% and **56%**. Energy and utilities are mostly sector (Exxon: 83%); McDonald's and Verizon are over 90% own risk.

4. What if a factor moves? A Monte Carlo scenario engine

The user picks a shock (e.g. VIX +50% in a month). Unshocked factors follow the conditional normal distribution given the shock; sectors respond through a joint regression on the six macro factors plus the market, with bootstrapped weekly residuals widened when the shock is unusually intense and with parameter uncertainty (50 bootstrap re-estimations); companies respond through their beta to their own sector ETF. 5,000 scenarios run live in about 0.05 s, and the web is coloured by the result.

Out-of-sample test (model fitted before each crisis)	Macro factors imposed	Macro factors and market imposed
Lehman 2008: predicted minus realised sector loss, average	+17.7 pts	+0.5 pts
COVID-19 2020	+16.8 pts	+3.5 pts
Rate hikes 2022	+10.4 pts	−7.7 pts
Sector-crises inside the 90% model interval (≈28 of 31 expected)	7 of 31	16 of 31

Reading: given the market move, the engine places sectors well; from macro factors alone it understates crises by about 15 points, because the factor–market link estimated in calm years misses panic. The 90% intervals are **model intervals, not confidence intervals**: in crises they are too narrow, and the cockpit shows the measured coverage next to every result.

5. Portfolio tools

- **Any portfolio** of the 110 companies, the ETFs, SPY or any Yahoo ticker (sector assigned automatically): Euler contributions to volatility and CVaR, 1-day, 10-day and 1-month forecasts with their own backtest, historical stress tests (2008, 2020, 2022) and the scenario engine's result for the portfolio.
- **Hierarchical Risk Parity** on the joint-fall distance, with CVaR as the cluster risk. Re-estimated every 63 days on the previous 504 only (out of sample), on an example portfolio: CVaR 2.56% against 3.47% for the user's weights and 2.95% for equal weights.
- **0–100 sector scorecard:** tail, systemic, stress, regime and momentum; the tail compares each sector both with the others and with its own history; Dirichlet draws of the weights give each sector a rank interval (XLK is first under every draw; ranks 2–4 are not distinguishable).

6. Engineering

Prefect flow with one task per step (collector, validator, models, backtests, scenarios, alerts, weekly report, publisher), scheduled with launchd every trading day at 22:30 Lisbon time on the NYSE calendar. Alerts and a weekly report go to an Obsidian vault; a weekly snapshot is published to a separate Git branch so the main branch holds only code. 206 pytest tests, each new calculation checked against an independent implementation, GitHub Actions CI.

Limitations

US sector ETFs and their largest holdings only. Several results rest on few crisis days (about 38 tail days per pair over 3 years). The scenario engine is linear and understates crises without the market move. HRP and the company ES correction are evaluated on relatively short or in-sample periods. It is an academic project, not investment advice.

Full methodology, formulas and every validation, in Portuguese: `docs/nota-metodologica.pdf` *(19 sections).*